unified-plx-document-knowledge-architecture

Unified PLX Document and Knowledge Architecture

Executive architecture report · `TASK-477` · 2026-08-04

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Executive decision

PLX should adopt **Knowledge Ledger with Disposable Projections**.

Business documents and records remain authoritative in SharePoint/Microsoft 365; customer, product, and vendor workflow state remains authoritative in Portal gold tables; technical content remains authoritative in Git. SharePoint is the system of record for Mission Control planning records, while Mission Control is workflow authority for execution, routing, and evidence policy. DocuSign remains authoritative for envelope execution evidence; registries remain authoritative for identities, tools, and skills.

A Postgres-first bitemporal ledger records versioned assertions, provenance, ontology releases, ingestion/projection receipts, and tombstones. It does not become another editable document store. Graph, vector, full-text, cache, and PLX Second-Brain surfaces are derived, version-stamped, security-trimmed, and rebuildable.

Why this architecture

PLX needs cross-system discovery and explanation without weakening ownership. The selected pattern:

- identifies one canonical source and authoring return path for every object;
- reconstructs what was believed, when it was valid, and when PLX learned it;
- exposes source, revision, owner, derivation, confidence, and conflicts;
- lets administrators evolve business vocabulary safely in a Portal workbench;
- detects duplicate, stale, and contradictory content without deleting it;
- keeps customer/vendor and source ACL boundaries ahead of retrieval;
- maps retention, privacy, residency, and Part 11 controls to explicit owners;
- lets department heads remain accountable while teams and agents execute within bounded, revocable autonomy;
- turns the roadmap into schema-validated milestones, dependencies, tasks, risks, gates, and evidence.

Architecture at a glance

```
flowchart LR
  sources["Canonical sources\nM365 · Portal · Git · MC · DocuSign · Registries"]
  ledger[("Versioned Knowledge Ledger\nassertions · provenance · ontology · receipts")]
  builders["Policy-aware projection builders"]
  projections["Disposable projections\ntext · vector · graph · cache · Second-Brain"]
  workbench["Portal\nsearch · evidence · ontology workbench"]

  sources -->|"versioned observations"| ledger
  ledger --> builders
  builders --> projections
  projections --> workbench
  workbench -->|"edit at source / governed proposal"| sources
```

The key control is the direction of authority: a search result or memory can lead back to its source; it cannot silently replace it.

Portal ontology workbench

Administrators and department stewards can:

1. propose a stable concept, label, synonym, mapping, hierarchy, or deprecation;
2. run structural and policy validation;
3. preview affected documents, assertions, searches, rules, skills, and agents;
4. collect department, Security, Quality, or Legal review based on impact;
5. publish an immutable, effective-dated ontology version;
6. rebuild only affected projections and inspect quality changes;
7. roll back by publishing a new attributable version.

Direct database/RDF editing and self-approval of high-impact changes are outside the operating model.

Discovery and knowledge quality

Retrieval is deny by default. Tenant, actor, source ACL, classification, purpose, retention, and policy filters are applied before ranking, cache reuse, or generation. Answers display canonical citations, revision time, derivation, confidence, and unresolved conflicts.

Duplicate and stale-document controls are non-destructive:

- exact IDs and hashes identify byte/source duplicates;
- normalized hashes and similarity identify near duplicates;
- semantic similarity proposes conceptual duplicates;
- source version and authority distinguish revision from duplication;
- review queues route candidates to the accountable department owner.

Staleness is source-aware: an old approved record can be current, while a recent copy can already be stale. Permission freshness fails closed.

Records, privacy, residency, and Part 11

Purview remains the source for M365 retention, record labels, legal holds, disposition, and proof of disposition. The ledger stores policy references and receipts; derived stores cannot outlive approved source policy or cross unapproved regions.

For records that Quality and Legal place inside the validated-system boundary, the design traces:

- validation for intended use;
- accurate and complete record copies;
- protection, retrieval, and approved retention;
- limited system access;
- secure, computer-generated, time-stamped audit trails that preserve prior information;
- operational and authority checks;
- training and accountability;
- signature name/date/time/meaning;
- signature uniqueness, identity verification, and record linkage;
- closed/open system classification and corresponding additional controls;
- applicable FDA electronic-signature certification evidence;
- non-biometric signature components, signing-session behavior, and genuine-owner controls.

This package is an architecture and validation plan, not a Part 11 compliance certificate.

Human and agent operating model

Department heads own meaning, risk, publication, and remediation. Quality and Legal own regulated-use and validation decisions. Security owns authorization, privacy, and residency approval.

Registered teams and agents may inventory, classify, propose, build disposable projections, execute tests, and draft evidence. They cannot decide regulatory scope, delete canonical records, weaken retention/access/residency, self-register capabilities, sign for a person, approve their own high-impact ontology change, or authorize live cutover.

Zero-drift program

The proposed Mission Control project is `PRJ-UNIFIED-KNOWLEDGE`. `TASK-477` should be re-homed without duplication. Seven buckets cover foundation, connectors, ontology, controls, discovery, autonomy, and rollout.

The committed program defines:

- 30 traced requirements;
- 8 dependency-ordered milestones;
- 20 tasks with owners, executors, autonomy bounds, dependencies, repositories, and done-when evidence;
- 10 risks;

- 8 approval gates;
- 5 validation commands.

`program.schema.json` freezes the shape. `scripts/check-unified-knowledge-program.py` validates schema, unique IDs, references, dependency DAGs, required invariants, milestone task coverage, and requirement task coverage. Its `--print-mc-plan` mode is deterministic and non-mutating.

Delivery sequence

Milestone	Outcome	Hard gate
MS-00	Estate, authority, compliance, privacy, authorization, and deployment facts	All Phase 0 decisions signed
MS-01	Bitemporal ledger and processing receipts	Integrity, backup, and recovery
MS-02	Canonical-source connectors	ACL, deletion, replay, deep-link, degradation
MS-03	Portal ontology workbench	Administrator usability and reversible publication
MS-04	Records, privacy, residency, and Part 11 controls	Quality/Legal/Security qualification
MS-05	Rebuildable projections and semantic discovery	Rebuild, quality, explanation, and leakage benchmarks
MS-06	Bounded agent/human workflows	Capability, revocation, approval, and MC evidence
MS-07	Architecture pilot and cutover readiness	Recovery, rollback, SLO, and owner approvals

Phase 0 blockers

No implementation should proceed until:

1. the live Microsoft 365 estate, licenses, Purview features, regions, administrators, and Graph permissions are evidenced;
2. pinned Portal and PLX Second-Brain repositories and runtime contracts are accessible;
3. Quality and Legal approve Part 11 intended use and the validated-system boundary;
4. identity, tenant isolation, privacy, purpose, and authorization policy is approved;
5. ledger/audit-store hosting, encryption, backups, recovery, residency, and bitemporal semantics are approved.
6. the recurring mirror-is-boring gate reports live, fresh data with `boringGateMet` true before ontology, discovery, or another new plane starts.

The Portal repository was not readable by this Cloud Agent, so the recorded decision to reuse Ricardo's Portal document stack is preserved but not misrepresented as independently verified.

Pilot

Use the existing Git-authoritative architecture collection first. It already declares generated-consumer status, provenance, canonical links, authoring return paths, and degraded behavior when Second-Brain is unavailable.

Pilot success requires complete provenance, correct authority recovery by technical and nontechnical users, no ACL/tenant leakage, visible staleness and conflict, usable canonical links during projection outages, zero-state projection rebuilds, deletion/ACL propagation, ontology rollback, and Mission Control evidence for every action.

Rollback

Disable the affected connector, capability, or projection; return users to canonical links and the prior approved projection; publish ontology rollback as a new version; rebuild from the last approved ledger watermark; preserve all source, ledger, audit, validation, and incident evidence; reopen the controlling Mission Control task before resuming.

Rollback never rewrites canonical documents or deletes ledger history.

Package

- `index.md` — canonical entry point
- `REQUIREMENTS.md` — requirement and acceptance contract
- `RESEARCH.md` — recovered evidence, independent source research, alternatives
- `SPEC.md` — detailed architecture and rollout specification
- `DECISIONS.md` — approved decisions and unresolved gates
- `REPORT.md` — this executive report
- `program.json` / `program.schema.json` — zero-drift execution contract
- PDF / DOCX — generated, shareable consumers of this report